

EUGENE L. BUCHANAN

Staff AI Engineer — Multi-Agent Frameworks & Agentic Infrastructure

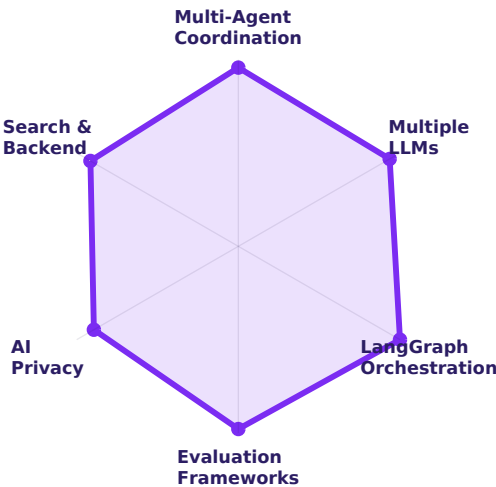
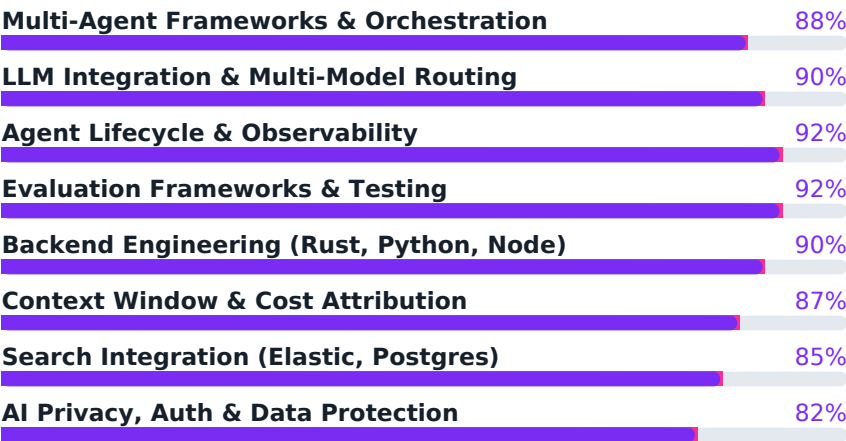


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Professional Summary

Staff AI engineer with 27 years in software and systems, focused on agentic AI infrastructure and multi-agent orchestration. I've shipped three production PRs to ZeroClaw (32.2k★ Rust gateway): a merged context-window meter for 9 LLM providers, a live provider-identity PR for usage events (cost attribution), and an agent lifecycle observer over JSON-RPC. I've built test automation, integration platforms, and CI/CD pipelines on AWS/Postgres/React. I focus on automatic model routing and platforms that deploy intelligent agents at scale.

Technical Expertise



Technology Stack

- LangGraph Multi-Agent Coordination Agent Lifecycle JSON-RPC Bounded I/O Fire-and-Forget
- Unix Domain Sockets
- Context Window Routing Multi-Model Cost Attribution MCP (Model Context Protocol) herdr-mcp Server
- External Tool Access Prompt Engineering OpenAI Anthropic Cohere Gemini
- Rust Python Node.js PostgreSQL AWS (ECS, S3) REST GraphQL Docker Git
- Cypress Selenium JUnit Postman Elasticsearch Regression Suites Health Dashboards Alerting
- SAML OAuth OIDC Permission Profiles Principal Isolation Deny-by-Default

Experience

Staff AI Engineer / Agentic Infrastructure Contributor (Open-Source)

2026 – Present

ZeroClaw Labs (github.com/zeroclaw-labs/zeroclaw) · Rust · 32.2k stars

- PR #7946 (MERGED) — context-window meter bar across TUI/gateway/CLI (+1,109 / -22, 25 files). One context_window source of truth for 9 providers; auto-populated config, doctor command, gateway API, live ContextBar. Removed drift between surfaces — directly relevant to Brain2's automatic model routing.
- PR #8966 (OPEN) — carry live provider identity on usage events (+2,742 / -141, 20 files). Fixed frozen 32k meter on 1M-token models; separated trim budget from model capacity; unconditional Usage events with serving-provider identity; per-provider usage breakdown on done frame — cost-attribution infra for multi-LLM orchestration.
- PR #8337 (OPEN) — herdr agent reporting integration (+1,712 / -17, 13 files). Observer reports agent lifecycle (idle/working/blocked/released) via JSON-RPC over UDS. Zero-config env detection, approval-gate blocked state, crash recovery via pane.release_agent. Multi-agent coordination infrastructure; maps to Super Agents' human-agent collaboration.

Senior Software Engineer / QA Lead / Integrations Product Owner

Dec 2020 – Aug 2025

Knowledgecity LLC · Remote · AWS / Postgres / React

- Product Owner, Integrations: shipped SAP, Oracle, Workday, Coursera, UKG, and Zoom integrations via SAML, OAuth, SFTP, and custom REST APIs — orchestrating multiple services with different auth models, data formats, and error semantics. Same coordination challenge as multi-agent systems.
- Built and maintained CI/CD quality pipelines with Cypress, Selenium, Postman, JUnit, Elastic, S3 — an evaluation framework for testing complex deployment scenarios across integration, backend, API, database, and frontend surfaces.
- Applied AI prompt engineering and shipped LLM-powered chatbot tooling in production, integrating it into customer-facing assistance and internal automation workflows.
- Designed evaluation frameworks, regression suites, health dashboards, and alerting to measure system-level dynamics across multi-environment deployments.
- Mentored and trained the test automation team; ran code reviews; raised engineering standards.

Senior Software Engineer II / Presales Engineering

1999 – 2001

RealNetworks · Seattle, WA · Streaming Media Platform

- QA on an advanced research team evaluating distributed system behavior across consumer appliances, mobile platforms, stream servers, and cellular networks — complex, multi-component scenarios.
- Global presales engineering: turned product capabilities into customer solutions across diverse technical environments. Trained new hires; led cross-functional collaboration.

Senior Software Test Engineer IV

1997 – 1999

Microsoft · Redmond, WA · Windows Media Server

- White-box testing and automation for Windows Media Server, Windows 98/NT — evaluation frameworks for complex system testing.
- Deep debugging across server infrastructure, network protocols, and media codecs.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International

Key Strengths

- **Multi-agent orchestration.** Agent lifecycle observer (PR #8337) + MCP server in 32k-star Rust gateway — shipped, not demoed.
- **Multi-model routing & cost.** Merged context-window meter for 9 LLMs (PR #7946); open PR adds live provider identity on usage events (PR #8966).
- **Evaluation frameworks.** 27 years test automation, regression suites, quality pipelines for complex distributed systems.
- **LLM integration.** Prompt engineering + chatbot tooling in production; orchestrates multiple providers daily.
- **Cross-functional collaboration.** Product Owner, QA Lead, presales engineer — works with PMs, designers, and engineers.